

Name: Bosire Meshack Nyagwencha
Neptun Code: IVF370

Task: CP2, Phase 1

A company stores the age and salaries of its employees.

Write a program that gives us how many different aged people are employed.

Specification:

In: $n \in \mathbb{N}, a \in \mathbb{N}[1..n]$

Out: $\text{cnt} \in \mathbb{Z}$

Fn: $\text{hm}: \mathbb{N} \rightarrow \mathbb{N}, \text{hm}(\text{ind}) = \text{COUNT}(j=1..n, a[\text{ind}] = a[j])$

Pre: $1 \leq n < 100$ and $\forall i \in [1..n]: (1 \leq a[i] \leq 100)$

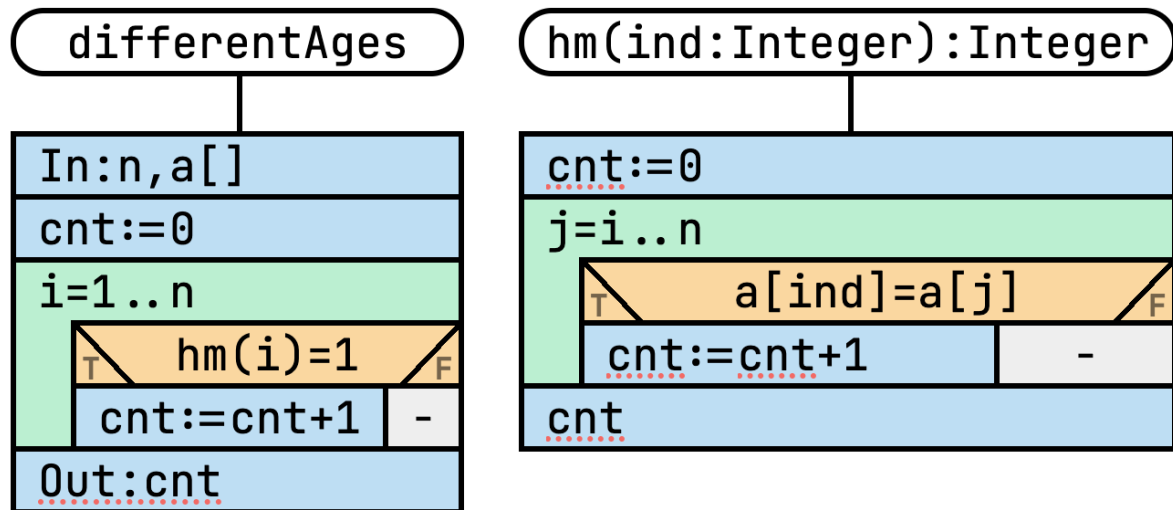
Post: $\text{cnt} = \text{COUNT}(i=1..n, \text{hm}(i) = 1)$

Pattern: counting

Algorithm:

differentAges

Pattern: COUNT b..e cnt A(i)	Task 1..n cnt hm(i)=1
hm function Pattern: COUNT b..e cnt A(i)	Task i..n cnt a[ind]=a[j]



Test cases:

1. invalid input: n:6 ; a: [-2,4,101] -> invalid..
2. no unique age: n:4 ; a :[20,40,20,40]->cnt=0
3. Count of different ages is 1 : n:5; a:[21,22,22,21,30]->cnt=1
4. 4 ages are unique...: n:6; a:[42,33,25,66,25,42]->cnt=4
5. all ages are unique... n:5; a:[31,32,45,47,50]->cnt=4